

A Sentiment-Based Comprehensive Rating Model with Extensive Dataset for Nationwide Hospital Rating in Bangladesh Using Natural Language Processing and Machine Learning

Asiful Kanza Auishik

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

asiful.kanza.auishik@g.bracu.ac.bd

Fariha Mohammed Al-Zahir

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

fariha.mohammed.alzahir@g.bracu.ac.bd

Magferah Sultana Samia

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

magferah.sultana.samia@g.bracu.ac.bd

Moumita Hossain

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

moumita.hossain@g.bracu.ac.bd

Abstract—Bangladesh has a huge number of public and private hospitals in its districts. However, there is no trustworthy, unbiased resource for patients to choose the best hospitals concerning quality of service. The existing star-based rating systems are prone to manipulation and do not capture detailed feedback from the patients. This paper proposes an advanced model of hospital rating that grades the hospitals based on online reviews of the patients, considering aspects like patient experience and quality of care. This proposed model employs NLP and ML techniques to analyze the sentiment of patient feedback and extract insights from it. It is expected to provide a data-driven hospital rating system based solely on user experiences by integrating various dimensions of hospital service quality to identify strengths and areas for improvement across the country. A large dataset of structured and unstructured reviews is collected from online platforms. Text mining and advanced NLP techniques process sentiment data. Various machine learning models, such as SVM, BERT, and CNN, are trained and validated on pre-processed data for sentiment prediction. Steps to achieving this objective involve data collection, data preprocessing, sentiment analysis, and, eventually, aspect-based sentiment analysis using zero-shot aspect detection to generate hospital ratings based on 4 aspects: treatment quality, cleanliness, affordability, and service quality. Rating generation classifies sentiments as positive, negative, or neutral, thereby dynamically, and in real time, rating a hospital. This model provides a more reliable and nuanced rating system that allows for transparent comparisons across hospitals and actionable insights into strengths and weaknesses. The framework is adaptable to other sectors, such as education and retail, providing an enlarged scope of application for sentiment analysis in service quality evaluation. For sentiment prediction of the reviews, BERT proves to be the best performing model out of all the models in terms of accuracy, precision, recall, and F1 score, producing 94.1%, 94.7%, 94.1%, and 94.4%, respectively. For the aspect-based ranking of hospitals, the model is most confident in detecting treatment quality as it produces the highest teacher threshold of 0.51 for this aspect. It also produces the

highest precision, F1 score, and agreement of 0.98, 0.97, and 0.96, respectively, for treatment quality, whereas, both affordability and service quality score the highest recall of 0.99.

Index Terms—Sentiment Analysis (SA), Advanced Hospital Rating Systems, Natural Language Processing (NLP), Machine Learning (ML), Patient Feedback, Google Reviews, Data-driven Rating, BERT (Bidirectional Encoder Representations from Transformers), Deep Learning (DL), Deep Neural Network (DNN), Convolutional Neural Network (CNN), Support Vector Machine (SVM), Text Mining Algorithms, Scalable Framework

I. INTRODUCTION

In recent years, online reviews have become a crucial source of feedback for assessing the quality of products and services across various sectors. In the healthcare industry, particularly in many developed countries, review systems have gained significant traction, enabling patients to make informed decisions based on quality metrics and ratings derived from reviews. Structured platforms, such as Statista® in Europe, provide detailed hospital rankings based on multiple performance metrics [1]. However, in Bangladesh, such resources remain largely underutilized. While patients occasionally share their experiences through comments and images on online platforms, these reviews are often disorganized and scattered, making it difficult to identify high-quality healthcare providers.

A pressing concern in Bangladesh's healthcare sector is the growing trend of patients seeking medical treatment abroad. Recent estimates suggest that Bangladeshis spend over \$4 billion annually on outbound healthcare tourism [2]. This trend reflects a lack of confidence in local medical services, as patients struggle to identify hospitals that excel in critical areas such as specialized services, cleanliness, time management,

surgical expertise, safety, and physician quality. While platforms like Google Maps provide basic star ratings, they often fail to capture the nuanced emotional insights embedded in patient reviews, leading to incomplete or misleading assessments of hospital performance. Additionally, some reviews serve as promotional content for specific hospitals, further complicating the evaluation process.

To address these challenges, developing a comprehensive hospital rating model requires constructing a robust dataset of categorized patient reviews that accurately reflect sentiments about healthcare facilities across Bangladesh. This study focuses on creating an extensive, well-organized dataset by aggregating data from online sources, particularly Google reviews, and developing a customized sentiment analysis model that combines existing models with adaptations tailored to our dataset. Specifically, we employ zero-shot learning to extract key aspects such as cleanliness, affordability, treatment quality, and service quality from patient reviews. These aspects are assigned weights to compute an overall sentiment score, which is used to rank hospitals. Our approach includes generating rankings for the top three hospitals overall and in each of the aforementioned aspects. The primary objective is to build a reliable database and evaluate various sentiment analysis (SA) models to identify the most accurate approach for generating an honest and representative hospital rating system for Bangladesh's healthcare sector.

A. Contributions

The contributions of this study are as follows:

- Development of a comprehensive dataset of patient reviews from online sources, specifically Google reviews, to support sentiment analysis for hospital ratings in Bangladesh.
- Design and implementation of a customized sentiment analysis model using NLP and ML techniques, incorporating zero-shot learning to extract key aspects (cleanliness, affordability, treatment quality, and service quality) and assigning weights to compute overall sentiment scores.
- Generation of hospital rankings, including the top three hospitals overall and in each extracted aspect, to provide a nuanced evaluation of healthcare facilities.
- Comparative analysis of multiple sentiment analysis models to determine the most effective approach for capturing patient sentiments and improving healthcare decision-making in Bangladesh.
- Creation of a structured methodology for data collection and preprocessing, ensuring the dataset is well-organized and suitable for advanced NLP-based sentiment analysis.

II. RELATED WORK

In this paper [3], the authors comprehensively compared the different approaches in multilingual SA (corpus-based, lexicon-based, and hybrid) and outlined the advantages and disadvantages of these approaches. Similar to the previous paper [3], this paper [4] studied Five Python and R libraries -

NLTK, TextBlob, Vader, Transformers (GPT and BERT pre-trained), and TidyText. Additionally, the author implemented four machine learning models - Tree of Decisions (DT), Support Vector Machine (SVM), Naive Bayes (NB), and K-Nearest Neighbor (KNN) in the study, applying sentiment analysis techniques. Another article [5] proposed supervised machine learning and an extended sentiment dictionary to perform Chinese sentiment analysis. The Naive Bayes classifier is used, resulting in an average precision score of 0.861, an average recall score of 0.843, and an average F1 score of 0.853. This paper [6] explored Deep Learning in the field of SA. DNN, CNN, and RNN algorithms were used to perform sentiment analysis on available datasets to determine their pros and cons. Similar to the paper [6], this paper [7] used DL and additionally implemented ML to perform a multi-class sentiment analysis in Urdu text. Two publicly available datasets, UCSA-21 and UCSA [8], were used to train the models. The authors' proposed mBERT classifier achieved the most promising result. Another research [9] on Urdu text was conducted using a hybrid dependency-based approach. The author reported the results using SVM, Logistic Regression, Multilayer Perceptron (MLP), and Decision Tree (DT) models, as well as DL models combined with dependency-based rules. The author summarized by stating that the hybrid approach outperformed the standard state-of-the-art SA method by nearly 10%. This paper [10] focuses on Natural Language Processing (NLP) and its ability to extract meaningful information more effectively. Using NLP, the authors were able to identify positive, negative, and neutral comments from a patient's experience feedback dataset with a weighted average precision, recall, and F1-score of 0.83, 0.82, and 0.82, respectively. In another paper [11], the authors used Supervised Machine Learning with an Extended Lexicon Dictionary to analyze the sentiment in Bangla text. Another paper [12] focused on ML and Lexicon-based approaches in SA and the comparison of the approaches. While a detailed comparison has been presented by the author in his study, the author concluded the experiment by stating that the supervised ML approach yielded 0.85 accuracy in his research. Similar to the paper [11], this paper [13] also focuses on Bangla text, in the context of the Ukraine-Russia war, but instead of a normal approach where ML and DL are dominantly used, the authors of this article trained and used BERT transformer models, namely multilingual BERT (mBERT), Distil-mBERT, XLM-RoBERTa, XLM-RoBERTa-large, BanglaBERT, to explore the outcome of the different approach. BanglaBERT outperforms all baseline and other transformer-based classifiers. This paper [14] comparatively studies SA using NLP and ML on US Airline Tweet data on Kaggle as a dataset, named Twitter US Airline Sentiment [15]. Another paper [16] researched a BERT-CNN-based approach in SA on an IMDB movie review dataset [17]. The author proposed a deep learning model in combination with BERT and CNN and compared the results with the sole BERT model. The result determines that the BERT-CNN model does not surpass the pure BERT model in terms of accuracy (BERT-CNN accuracy 0.929 and

BERT accuracy 0.930). Two more papers [18] [19] studied different TRANSFORMER based language models and made a comparison, outlining the results. The latter paper did Bangla Sentiment Analysis using Weighted and Majority Voted Fine-Tuned Transformers. Amongst all the models, DistillBERT and BanglaBERT performed the best. This paper [20] proposed a different idea of foreign language sentiment analysis by translating those languages into a base language, English, and then applying sentiment analysis. The text translation was performed using LibreTranslate and Google Translate, and sentiment analysis was performed using Twitter-Roberta-Base, Bertweet-Base, GPT-3, and a novel Proposed Ensemble model. Amongst these, Google Translate, with the Proposed Ensemble model, achieved the overall highest scores across all metrics. Paper [21] evaluates multilingual sentiment analysis techniques for under-resourced languages, emphasizing that deep learning models like CNN and RNN are used in more than 60% of the studies. The paper identifies the drawbacks of each technique used in sentiment analysis and concludes that a combination of deep learning models, such as CNN, Bi-LSTM, and LSTM, can significantly enhance multilingual sentiment analysis and proposes a DL learning framework independent of any MT-based methods. For the extraction of six individual emotions - happy, sad, tender, excited, angry, and scared - from Bangla text for both article and sentence, paper [22] implements two machine learning techniques, Naive Bayes and Topical approach, where the Topical approach surpasses Naive Bayes. Paper [23] compares the semantic stability of BERT and Word2Vec across four time spans (3, 5, 10, and 20 years) using 20 years of People’s Daily articles, concluding that BERT maintains stronger overall semantic stability over both short and long durations due to scoring higher SD, MTS, and LNS and lower RSC in every period compared to Word2Vec.

III. METHODOLOGY

A. Dataset Description

The dataset consists of reviews of nationwide hospitals extracted from online Google Maps reviews. Real-world data often contains flaws like punctuation, missing values, incorrect translations, and so on. If they are not processed before training, the model may learn inaccurate patterns, producing less accurate results. Therefore, the following preprocessing steps have been applied to achieve the best results.

B. Data Preprocessing

The preprocessing pipeline consists of several essential steps to clean and prepare the raw review data for model training. Algorithm 1 presents the pseudocode for the complete preprocessing workflow.

The preprocessing steps include: (1) Removing emojis to eliminate non-textual symbols, (2) Removing blank rows to ensure data quality, (3) Converting text to lowercase and removing punctuation marks to standardize the input, and (4) Tokenization and text vectorization to transform text into numerical representations suitable for machine learning models.

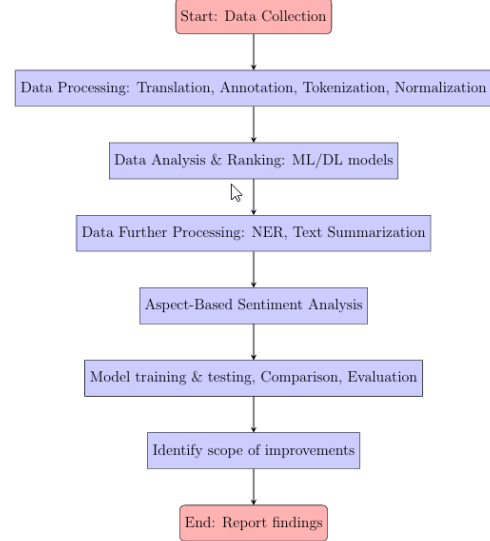


Figure 2.1: Work Plan Flowchart

Fig. 1: Workflow diagram of the proposed hospital rating system.

Algorithm 1 Data Preprocessing Pipeline

```

1: Input: Raw review dataset  $D$ 
2: Output: Preprocessed dataset  $D'$ 
3:
4: for each review  $r$  in  $D$  do
5:    $r \leftarrow \text{RemoveEmojis}(r)$ 
6:   if  $r$  is blank then
7:     Remove  $r$  from  $D$ 
8:   continue
9: end if
10:  $r \leftarrow \text{ConvertToLowerCase}(r)$ 
11:  $r \leftarrow \text{RemovePunctuation}(r)$ 
12:  $tokens \leftarrow \text{Tokenize}(r)$ 
13:  $vectors \leftarrow \text{VectorizeText}(tokens)$ 
14: Add  $vectors$  to  $D'$ 
15: end for
16: return  $D'$ 
  
```

C. Model Descriptions

The dataset was used to train five machine learning models and two deep learning models. Each model has distinct characteristics suited for different aspects of sentiment analysis:

Decision Tree Classifier: A tree-based model that splits data based on feature values to make predictions. It is interpretable and handles non-linear relationships well but can overfit on complex datasets. Its strength lies in visualization and understanding decision boundaries.

Support Vector Machine (SVM): A robust classifier that finds the optimal hyperplane separating different sentiment classes. SVM performs well in high-dimensional spaces and

is effective with clear margin separation. However, it requires careful kernel selection and can be computationally expensive for large datasets.

Logistic Regression: A linear model that estimates the probability of sentiment classes using a logistic function. It is simple, fast, and provides probabilistic outputs. While efficient for linearly separable data, it may underperform on complex non-linear patterns.

Random Forest: An ensemble method combining multiple decision trees to improve prediction accuracy and reduce overfitting. It handles feature interactions well and provides feature importance rankings. The model is robust but can be memory-intensive and slower to train.

Gaussian Naive Bayes: A probabilistic classifier based on Bayes' theorem assuming feature independence. It is computationally efficient and works well with small datasets. However, the independence assumption may not hold for correlated features in text data.

BERT (Bidirectional Encoder Representations from Transformers): A state-of-the-art transformer-based model pre-trained on large text corpora. BERT captures bidirectional context and semantic relationships effectively, making it highly accurate for NLP tasks. Its main limitation is computational cost and resource requirements.

CNN (Convolutional Neural Network): A deep learning architecture originally designed for image processing but adapted for text classification. CNNs extract local features through convolutional filters and capture n-gram patterns effectively. They are faster than transformers but may miss long-range dependencies.

Zero-shot and Sentence-level Sentiment: The suggested aspect-based sentiment analysis approach utilizes a multilingual zero-shot teacher model, joeddav/xlm-roberta-large-xnli, to extract target service aspects from hospital review sentences directly without needing annotated data. The model defines aspect detection as a natural language inference task by evaluating four hypotheses - cleanliness, affordability, quality of treatment, and quality of service; and outputting for each a probability score, reflecting how probable a sentence is describing the respective aspect. This sentence-based strategy minimizes topic dilution while maximizing detection accuracy within multilingual, code-mixed environments.

For enhanced scalability, a student classifier distilled from xlm-roberta-base is learned from the teacher's soft labels. The student classifies aspect probabilities in a single pass with a multi-label binary cross-entropy objective with weighted samples, with considerable saving in inference time at high teacher output fidelity.

In tandem, sentiment is classified for every sentence with cardiffnlp/twitter-xlm-roberta-base-sentiment, mapping polarity onto -1 (negative), 0 (neutral), or $+1$ (positive) scores so aspect mentions are associated with their corresponding sentiment orientation.

For strong thresholding and calibration, both teacher and student model raw probabilities to binary aspect-presence tags with data-driven cut-offs optimized using Otsu's algorithm.

This aligns decision boundaries with score distributions to allow consistent cross-aspect interpretation without manually defined thresholds.

The complete aspect detection pipeline combines these components sequentially. A sentence is fed through the distilled student and zero-shot teacher first to generate aspect probability predictions. These are then combined with sentiment predictions and per-aspect thresholding to assign polarity-tagged aspect mentions. At the review level, aspect-specific sentiment scores are averaged only over sentences with that aspect to avoid leaking sentiment from irrelevant text. This multi-stage pipeline sacrifices accuracy, interpretability, and computational expense, and is intended to be large-scale-hospital-review-analysis-friendly.

D. Model Evaluation

The performance of each model was assessed using multiple evaluation metrics to ensure comprehensive analysis:

Accuracy Score: Measures the proportion of correctly classified instances out of total predictions, providing an overall performance indicator.

Precision: Calculates the ratio of true positive predictions to all positive predictions, indicating how many predicted positive sentiments are actually correct.

Recall: Computes the ratio of true positive predictions to all actual positive instances, showing the model's ability to identify all relevant positive cases.

F1 Score: Represents the harmonic mean of precision and recall, balancing both metrics to provide a single performance measure especially useful for imbalanced datasets.

Confusion Matrix: A tabular visualization showing true positives, true negatives, false positives, and false negatives, enabling detailed error analysis.

Word Cloud: A visual representation of frequently occurring words in different sentiment categories, helping identify key terms associated with positive, negative, and neutral sentiments.

Agreement: Quantifies the degree to which two models provide identical results for the same aspect.

Pearson Correlation Coefficient: Quantifies the linear relationship between two continuous variables.

IV. RESULTS AND DISCUSSION

A. Result Description

The experimental results reveal distinct performance patterns across different divisions of Bangladesh. For the Rangpur division, Random Forest achieved the highest accuracy (0.819) and recall (0.819), while BERT demonstrated superior precision (0.822) and F1 score (0.801). This suggests that Random Forest effectively captured the overall patterns in Rangpur's data, but BERT's contextual understanding provided more precise positive sentiment identification.

BERT consistently outperformed other models across Rajshahi, Dhaka, and Sylhet divisions in all four metrics, demonstrating its robust capability to understand the semantic nuances in patient reviews. The exceptional performance in

Sylhet (accuracy: 0.941, precision: 0.947, recall: 0.941, F1: 0.944) indicates that reviews from this division contained clearer sentiment expressions with less ambiguity, allowing BERT’s attention mechanism to effectively capture sentiment patterns.

For Mymensingh division, while SVM achieved the highest precision (0.876), BERT surpassed it in accuracy (0.889), recall (0.889), and F1 score (0.873). This pattern suggests that although SVM was conservative in positive predictions, BERT achieved better overall balance. Similarly, in Barisal, Khulna, and Chittagong divisions, BERT maintained superior performance across all metrics, with SVM following closely in Khulna.

The dominance of BERT across most divisions can be attributed to several factors: (1) its pre-trained bidirectional context understanding captures subtle sentiment cues that traditional models miss, (2) the transformer architecture’s attention mechanism effectively weighs important words and phrases in reviews, (3) BERT’s ability to handle complex sentence structures and medical terminology common in healthcare reviews, and (4) the model’s capacity to understand context-dependent meanings where the same words may express different sentiments based on surrounding text.

The zero-shot teacher model, xlm-roberta-large-xnli, is distilled into a lighter student model, xlm-roberta-base, that predicts four aspect probabilities from a sentence in one forward pass. From Fig. 4, training loss decreases steadily from 0.256 at the start to 0.17 by the end of epoch 15, with only small minibatch-level wiggles. The curve shows the classic “fast early drop, slower later improvement,” and there’s no upward drift, evidence of stable optimization and no obvious overfitting.

Out of all the 4 aspects, treatment quality produces the highest Otsu threshold ($\tau = 0.51$), reflecting that the teacher model is most confident in detecting this aspect in reviews and that only confident, explicit mentions of treatment quality are classified as positive. On the other hand, service quality produces the lowest Otsu threshold ($\tau = 0.34$), capturing a broader range of moderate service-related mentions and compensating for the noisier linguistic variability in staff and behavior descriptions, and cleanliness and affordability produce Otsu thresholds of 0.39 and 0.41, respectively.

Graphs of teacher-student agreement with respect to threshold are produced. The student model produces the highest precision (0.98), F1 score (0.97), and agreement (0.96) for treatment quality, which indicates that it correctly identifies most mentions of treatment quality in hospital reviews while minimizing false positives. Both affordability and service quality surpass the other aspects in terms of recall (0.99), implying that the student captures almost all teacher-positives for both aspects. However, the student generates the lowest scores for service quality in terms of precision (0.64), F1 score (0.78), and agreement (0.84), indicating that there are relatively higher false positives and maximum difference in opinions of student and teacher regarding service quality.

Comparing teacher and student probabilities show the high-

est Pearson correlation coefficient (0.984) occurs for cleanliness, denoting the highest consistency between teacher and student probabilities for that aspect and that student learns the cleanliness related statements the most effectively. On the other hand, the weakest correlation (0.970) is shown for service quality which could be the consequence of lesser mentions of service quality in training samples and the aspect’s subjective linguistic variability in user feedback.

B. Figures and Tables

TABLE I: Performance Comparison of Models Across Divisions

Division	Model	Accuracy	Precision	Recall	F1 Score
Rangpur	Random Forest	0.819	0.799	0.819	0.798
	BERT	0.812	0.822	0.801	0.801
Rajshahi	BERT	0.930	0.928	0.931	0.929
Dhaka	BERT	0.866	0.865	0.866	0.864
Mymensingh	SVM	0.872	0.876	0.870	0.868
	BERT	0.889	0.869	0.889	0.873
Sylhet	BERT	0.941	0.947	0.941	0.944
Barisal	BERT	0.825	0.819	0.825	0.818
Khulna	BERT	0.908	0.910	0.908	0.908
	SVM	0.902	0.907	0.902	0.901
Chittagong	BERT	0.841	0.824	0.841	0.830

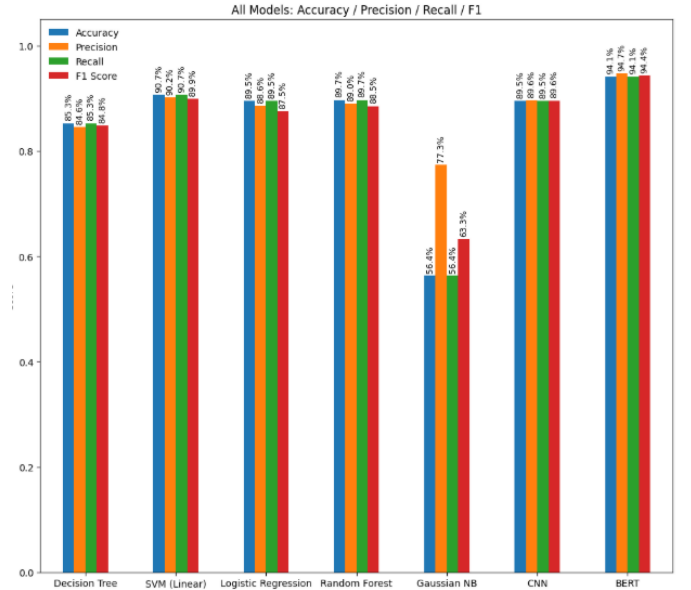


Fig. 2: Performance comparison of different models for Sylhet division showing BERT’s superior performance across all metrics.

C. Comparative Result Analysis

A comparative analysis of model performance across the 8 divisions reveals notable regional variations in sentiment-based hospital rating predictions. Among all divisions, Sylhet demonstrated the highest model performance with BERT achieving 94.1% accuracy. This indicates that sentiment cues

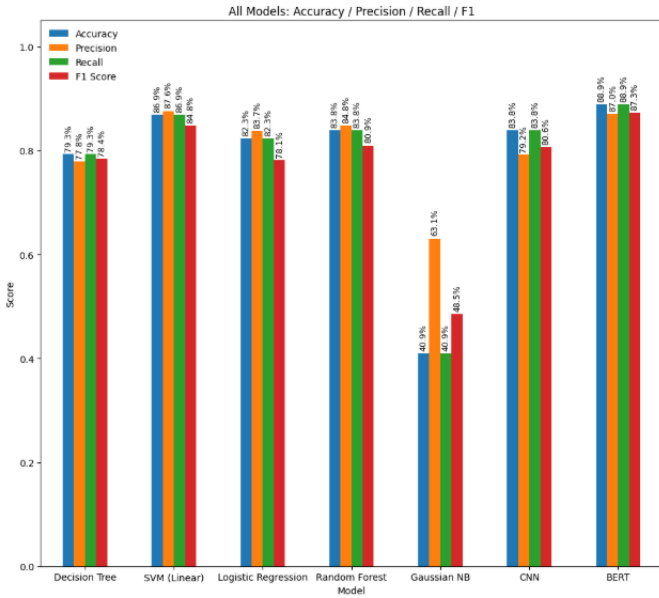


Fig. 3: Performance comparison of different models for Mymensingh division illustrating competitive results between BERT and SVM.

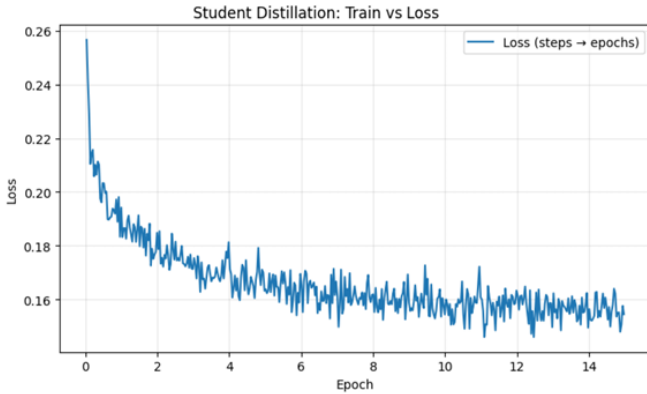


Fig. 4: Student model training loss over fifteen epochs during distillation for Dhaka hospital reviews.

in Sylhet’s hospital reviews were more consistent or better aligned with ground truth ratings, potentially due to clear review expressions and fewer neutral sentiments. The result metrics of the Rajshahi division also show a well-balanced dataset with clearly separable sentiment polarity, with BERT achieving 93.0% accuracy.

In contrast, Rangpur and Barisal recorded comparatively lower overall scores. For Rangpur, the best recall was achieved by Random Forest (0.819), but BERT still outperformed in terms of F1 score and precision, suggesting that while Random Forest captured more true positives, BERT maintained better precision-recall balance. The other divisions - Dhaka, Khulna, Chittagong and Mymensingh - show that patient feedback consisted of significant mixed sentiments rather than pure positive and negative sentiments, making classification more

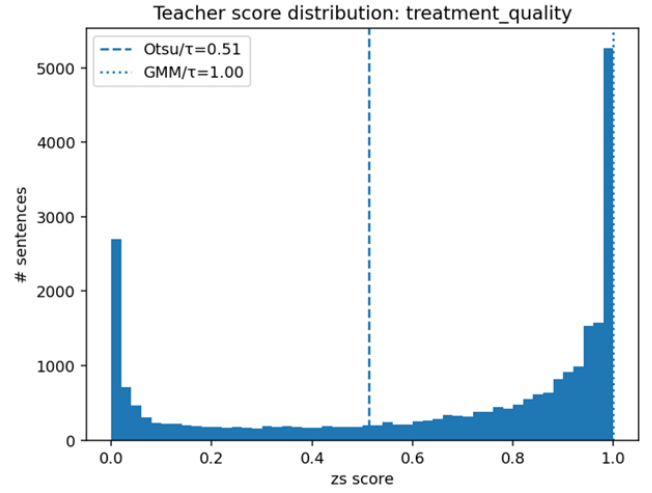


Fig. 5: Teacher score distribution for treatment quality.

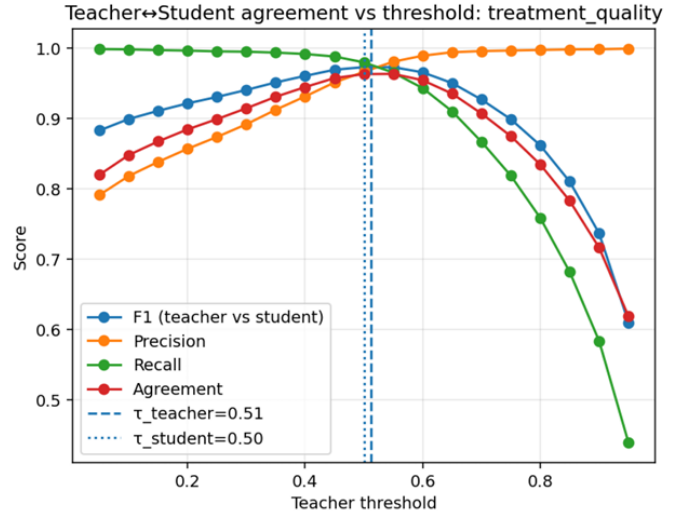


Fig. 6: Teacher–student agreement vs threshold for treatment quality.

challenging. However, BERT’s contextual understanding still enabled it to achieve the highest performance in most cases.

V. CHALLENGES AND LIMITATIONS

Throughout this research, several challenges and limitations were encountered that influenced the performance and generalizability of the proposed system.

Data Imbalance: The dataset showed a significant imbalance, with hospitals in Dhaka being heavily overrepresented compared to those in other regions of Bangladesh. This disproportion may cause the model to learn primarily from urban data, limiting its effectiveness in underrepresented areas. Although a confidence-score weighting mechanism was implemented to reduce this bias, complete correction was not achievable. As a result, hospitals with fewer reviews remained sensitive to individual review sentiments, potentially leading to unstable or skewed rankings.

Data Quality: Online review data often contained incomplete, inconsistent, or subjective expressions. Some reviews lacked context or included irrelevant details, which lowered the overall data reliability. Extensive preprocessing and validation were necessary to ensure that the remaining reviews accurately reflected genuine hospital experiences.

Sentiment Analysis Complexity: Understanding the nuances of human emotion posed a major challenge. The presence of sarcasm, implicit sentiment, domain-specific medical terms, and mixed opinions within a single review limited the model's accuracy in detecting true sentiment polarity. These linguistic complexities occasionally affected the precision of sentiment classification and, consequently, the reliability of the ranking process.

Review Bias: Online platforms inherently attract polarized opinions, as individuals with either very positive or very negative experiences are more likely to leave reviews. This self-selection bias can distort the sentiment distribution and misrepresent the average patient experience, undermining the fairness of the final ranking.

Limited Data Source: The dataset was collected solely from Google Maps reviews, excluding other social or offline feedback channels such as surveys or medical forums. This single-source dependency may not fully capture the multidimensional nature of patient experiences, limiting the comprehensiveness of hospital evaluation.

Computational Constraints: The zero-shot sentiment analysis model used in this study is computationally intensive, requiring high processing power and extended inference times. This restricts scalability and may hinder real-time deployment on resource-constrained systems.

Dynamic Update Limitation: The current implementation processes data in static batches. Although effective for initial evaluation, it lacks a dynamic pipeline to automatically integrate new reviews and update hospital rankings in real time. Future work will focus on developing an automated data ingestion and model update mechanism to maintain continuous accuracy in rankings.

Ethical and Privacy Considerations: Using publicly available patient feedback introduces ethical challenges related to privacy, data ownership, and consent. Despite many efforts, ensuring complete ethical compliance and user privacy remains an ongoing concern that future iterations must address through stricter data governance frameworks.

VI. CONCLUSION AND FUTURE WORKS

This paper presents a sentiment-based hospital rating framework that leverages machine learning and natural language processing (NLP) to evaluate patient feedback from hospitals across Bangladesh. The proposed system addresses the existing research gap by constructing a large-scale, publicly available hospital review dataset—the first of its kind in Bangladesh—covering over 27,000 reviews and 67,000 sentences across all eight divisions. This dataset serves as a foundation for transparent and data-driven healthcare quality assessment.

Multiple machine learning and deep learning models were implemented, including Decision Tree Classifier, Support Vector Machine (SVM), Logistic Regression, Random Forest, Gaussian Naïve Bayes, Convolutional Neural Network (CNN), and BERT. The experimental results demonstrate that BERT consistently outperformed all other models across all metrics. The Sylhet division achieved the highest performance, with a precision of 94.1.

Beyond review-level sentiment analysis, the most significant contribution of this research is the development of an aspect-based sentiment analysis (ABSA) framework. Rather than assigning a single overall rating to each hospital, the system generates separate sentiment scores for four critical dimensions: treatment quality, cleanliness, affordability, and service quality. This was achieved through a zero-shot learning approach using XLM-RoBERTa-Large as a teacher model, which identifies aspect mentions at the sentence level without requiring manual annotation. The knowledge from the teacher was distilled into a lightweight XLM-RoBERTa-Base student model, enabling efficient processing of 67,000 sentences while maintaining high accuracy (F10.9). This multi-dimensional ranking approach transforms hospital evaluation from simplistic star ratings into granular, actionable metrics that reflect patients' specific priorities. Patients can now identify hospitals excelling in particular domains—whether seeking affordable care, high-quality treatment, cleanliness, or superior service—enabling genuinely informed healthcare decisions tailored to individual needs. The aspect-based framework also provides hospitals with specific, categorized feedback on where improvements are needed, promoting targeted service enhancement.

The developed framework enables policymakers, healthcare administrators, and patients to make informed, data-supported decisions. It promotes accountability and transparency in the healthcare sector while offering a scalable foundation adaptable to other domains such as education, retail, and public service evaluation.

Future Works: Future improvements will focus on incorporating domain-specific sentiment analysis for more fine-grained hospital evaluations. A real-time data pipeline will be developed to integrate new patient reviews and update hospital rankings automatically and dynamically. Integrating static hospital metadata (e.g., number of beds, facilities, accreditation) with dynamic sentiment data is expected to produce a hybrid rating model for more comprehensive evaluation. Furthermore, multi-source data integration—including social media, medical forums, and offline survey feedback—will enhance model robustness. Advanced techniques, such as bias detection, explainable AI (XAI), and model optimization, will also be explored to ensure the reliability, interpretability, and scalability of the system in low-resource environments.

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